Usability Evaluation of World Model Studio

Anonymous Author(s)

***Abstract*—This paper presents a usability evaluation of World Model Studio, a research-stage application for generating and viewing world model outputs for XR and virtual production contexts. Evaluators familiar with XR, virtual production, 3D workflows, or related fields tested the main interface flow using Jakob Nielsen's 10 usability heuristics. The interface received a mean usability score of 38.42/50, placing it in the good usability range. Results show strengths in consistency, minimalist design, and recognition rather than recall, while highlighting the need for clearer system-status feedback, better error recovery, and advanced controls before broader production-oriented use. The study provides evidence-based recommendations for future interface development.**

***Index Terms*—heuristic evaluation, world models, XR, virtual production, Nielsen heuristics**

I. Introduction

World Model Studio is a research-stage application for generating and viewing 3D world representations. In this paper, world models refer to learned or generated 3D environments that support visual exploration, scene understanding, and interaction [1]. The application includes Gaussian-splat outputs for XR and virtual production workflows [2].

Virtual production depends on real-time visualization, game engines, VR/AR systems, camera tracking, LED volumes, and in-camera visual effects [4]. Since these workflows require scenes that can be created and adapted quickly, world-model-based tools may support faster concept development, previsualization, and early production planning.

World Model Studio allows users to select a generation mode, enter text or image inputs, generate outputs, and inspect results in browser-based viewers. Fig. 1 illustrates the evaluated interface, while Table I summarizes the highest-priority usability improvements identified from the results. This study evaluates the tool's usability for XR and virtual production workflows using Jakob Nielsen's 10 usability heuristics [3], focusing on interface evaluation rather than artistic quality, model training, or rendering performance.

II. Literature Review

Existing literature shows that virtual production is becoming more interactive and accessible beyond large studio environments. Swords and Willment [4] describe virtual production as a workflow combining real-time visualization, game engines, VR/AR systems, motion capture, camera tracking, LED volumes, and in-camera visual effects. Recent work also examines AI-generated environments for virtual film production [5]. However, these studies do not fully address whether a world-model-based interface can be used independently by XR and virtual production users. This paper addresses that gap through a usability evaluation of World Model Studio as a user-facing interface rather than only as a technical generation system.

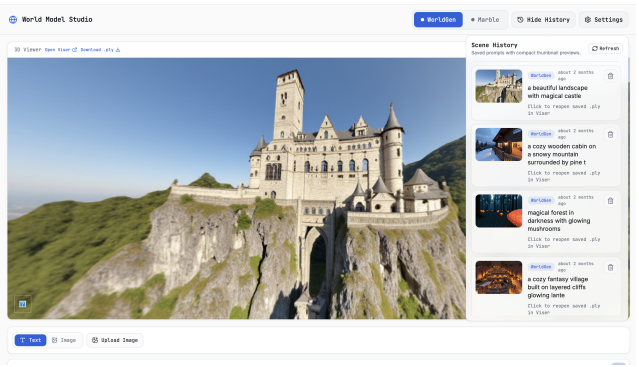


Fig. 1. World Model Studio interface with a generated scene displayed in the viewer.

III. Methodology

The evaluation involved 12 participants with experience in XR, virtual production, 3D workflows, or related fields. Participants were 21–50 years old, with an average age of 32.3. The gender distribution was eight male and four female participants. They reported approximately 2 to 23 years of relevant experience, with an average of about 8.8 years. Each evaluation session lasted 30–45 minutes, and all results were anonymized and reported in aggregated form.

The total usability score for each participant was calculated out of 50 by summing the 10 heuristic scores. Mean scores were also calculated for each heuristic to identify the strongest and weakest usability areas. Qualitative comments were grouped into recurring themes, and recommendations were organized by priority: high for issues that could block task completion, medium for issues causing confusion or delay, and low for minor refinements or optional improvements.

IV. Data Analysis and Results

World Model Studio received a combined usability score of 461/600. The mean score was 38.42/50, with a median of 40/50. According to the evaluation matrix, this places the interface in the good usability range. Six participants rated it as excellent, four as good, and two as showing moderate usability issues.

At the heuristic level, the strongest categories were recognition rather than recall (4.67/5), consistency and standards (4.42/5), and aesthetic and minimalist design (4.42/5). Participants described the interface as clean, minimal, and understandable after a short exploration period. The weakest categories were help and documentation (2.92/5), system-status visibility (3.25/5), error prevention (3.54/5), and flexibility and efficiency of use (3.58/5).

TABLE I
HIGH-PRIORITY USABILITY IMPROVEMENTS

Priority Area	**Main Improvement Focus**
System-status visibility	Progress and loading feedback
Help and onboarding	Tooltips and short onboarding support
User control and recovery	Cancel, reuse, and recovery options
Error prevention	Input guidance and warning messages

Table I summarizes the main improvement areas identified from the weakest heuristic scores. System-status visibility should be improved with clear progress indicators, loading messages, and generation-status feedback so users understand when generation or viewer loading is in progress. Help and onboarding should be strengthened through a short guide, contextual tooltips, and plain-language explanations of generation modes and controls. User control and recovery should be improved by adding cancel options, prompt/history reuse, clearer error messages, and recovery paths. Error prevention should be supported through input guidance, warnings before risky actions, and clearer constraints for text and image prompts.

V. DISCUSSION

The findings suggest that World Model Studio can support early XR and virtual production exploration, especially for concept development and visualization. The positive overall score indicates that participants could understand and use the main interface flow after a short exploration period. However, the lower scores for help, system-status visibility, error prevention, and flexibility show that the current interface still needs stronger guidance and recovery support before broader production-oriented use.

These results are especially relevant for XR and virtual production contexts, where users may need to iterate quickly, understand generation progress, and recover from failed or unclear outputs during creative work. Since this study focused on interface usability rather than output quality or rendering performance, the results should be interpreted as an early evaluation of the tool's interaction design rather than a complete assessment of the underlying world-model pipeline.

The results also suggest that usability improvements should be prioritized before evaluating production readiness. In particular, clearer status feedback and recovery options are important because XR and virtual production workflows often involve iterative decision-making, rapid scene testing, and repeated adjustments. Improving these interaction points may help users maintain confidence during generation and inspection tasks, especially when outputs take time to load or require revision.

VI. CONCLUSION AND FUTURE WORK

This paper evaluated World Model Studio using Nielsen's 10 usability heuristics. The interface received a positive overall score of 38.42/50, with strengths in recognition, consistency, and minimalist design. These results suggest that the tool has potential for early XR and virtual production exploration, but broader testing should follow after improving help documentation, system-status feedback, error prevention, and recovery support. Future work should evaluate an improved version with a larger and more diverse user group, include task-completion measures, and compare usability outcomes across different XR and virtual production scenarios.

ACKNOWLEDGMENT

The authors thank the evaluators who participated in the usability evaluation and provided valuable feedback.

REFERENCES

[1] D. Ha and J. Schmidhuber, "World Models," Mar. 2018, doi: 10.5281/zenodo.1207631.

[2] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian Splatting for Real-Time Radiance Field Rendering," *ACM Transactions on Graphics*, vol. 42, no. 4, Art. no. 139, 2023, doi: 10.1145/3592433.

[3] J. Nielsen, "10 Usability Heuristics for User Interface Design," Nielsen Norman Group. Accessed: May 12, 2026. [Online]. Available: https://www.nngroup.com/articles/ten-usability-heuristics/

[4] J. Swords and N. Willment, "The emergence of virtual production – a research agenda," *Convergence: The International Journal of Research into New Media Technologies*, vol. 30, no. 5, pp. 1557–1574, Oct. 2024, doi: 10.1177/13548565241253903.

[5] P. Leininger, C. J. Weber, and S. Rothe, "Understanding Creative Potential and Use Cases of AI-Generated Environments for Virtual Film Productions: Insights from Industry Professionals," in *Proceedings of the 2025 ACM International Conference on Interactive Media Experiences*, New York, NY, USA: ACM, May 2025, pp. 60–78, doi: 10.1145/3706370.3727853.